Algorithm & Programming Cheat Sheet

Discrete Math

Logic Propositions

Propositions are statements that can be TRUE (T) or FALSE (F).

Common Operators:

- NOT (NOT P): Negation (makes the opposite)
- AND (P AND Q): Both must be TRUE to get TRUE
- OR (P OR Q): At least one must be TRUE
- IF... THEN (IF P THEN Q): Only FALSE if P is TRUE and Q is FALSE
- IF AND ONLY IF (P IFF Q): TRUE if both are the same (both TRUE or both FALSE)

Truth Tables:

NOT (NOT P)

P | NOT P

T | F

F | T

AND (P AND Q)

P | Q | P AND Q

T | T | T

T | F | F

F | T | F

F | F | F

OR (P OR Q)

P | Q | P OR Q

T | T | T

T | F | T

F | T | T

F | F | F

IF... THEN (IF P THEN Q)

P | Q | IF P THEN Q

T | T | T

T | F | F

F | T | T

F | F | T

IF AND ONLY IF (P IFF Q)

P | Q | P IFF Q

T | T | T

T | F | F

F | T | F

F | F | T

Easy Examples:

- P: "I study."

- Q: "I pass the exam."

Example Propositions:

- NOT P: "I do not study."
- P AND Q: "I study AND I pass the exam."
- P OR Q: "I study OR I pass the exam."
- IF P THEN Q: "If I study, then I pass the exam."
- P IFF Q: "I study if and only if I pass the exam."

Semantic Tree

Used to check if a proposition is valid by breaking it down into all possible truth combinations.

Contradiction Proof

To prove a statement is TRUE by assuming the opposite and finding a contradiction.

Programming Basics

Algorithm

A clear and logical sequence of steps to solve a problem. It's not yet code, just the plan.

Program

The actual code that follows the algorithm.

Programming

The process of writing the program based on the algorithm.

Keep going, you're doing amazing!